Tiny Wins (Nốt)
Software Project

One positive note every day

PRM393 · Lab 3 + Final Assignment

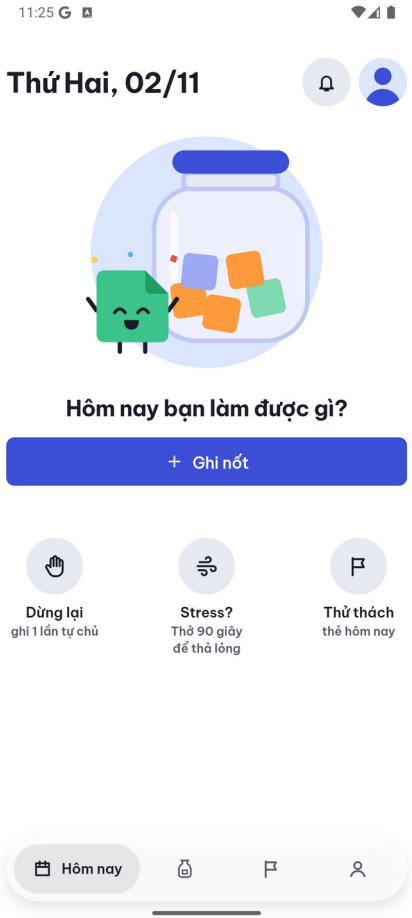
Đoàn Thị Kim Thúy · Phùng Thế Quân · Phan Đức Duy · Phan Hữu Phước

Introduction

Small wins, meaningful progress.

Nốt helps students record their moods and small wins, then revisit them in a digital Jar.

- **USER** — Students under academic pressure
- **PURPOSE** — Daily reflection and positive memories
- **TARGET** — Mood + one win in under 30 seconds



Context

Minh, 22 · Final-year student

After studying

Unfinished tasks still occupy his attention.

At the end of the day

Small achievements receive little attention.

Later

The positive moments are easy to forget.

Illustrative persona from the project design, not a survey respondent.

Problems

Three pain points identified from the project persona.

01 Unfinished work dominates

Attention stays on tasks left undone.

02 Small wins go unrecorded

There is no daily record to revisit.

03 Progress is hard to notice

Positive moments are scattered or forgotten.

Evidence status: Persona-based rationale; user interview or survey data has not been supplied.

Proposed Solution

A daily record of what went right.

PAIN POINT	NỐT FEATURE	EXPECTED VALUE
Small wins forgotten	**Daily Notes & Jar**	Record and revisit notes
No regular reflection	**Mood Tracking & Journal**	Reflect with a simple routine
Progress is hard to see	**Journey, Streaks & Weekly Card**	Review progress over time

Core: personal notes + challenge workflow

Extensions: AI, Discord, Admin Console

Main Actors

User · Mobile

Write notes, complete challenges, submit personal cards

Staff · Web Admin

Manage catalog, moderate submissions, post announcements

Admin · Web Admin

Staff capabilities + manage user roles (not their own)

Discord User · DM

Link an Nốt account and manage notes through messages

Main Features

Daily Notes & Jar

Moods, notes, journal and saved memories

Challenges & Moderation

Complete, submit and review challenge cards

Self-Control & Breathing

Check-ins and guided breathing

Journey & Weekly Card

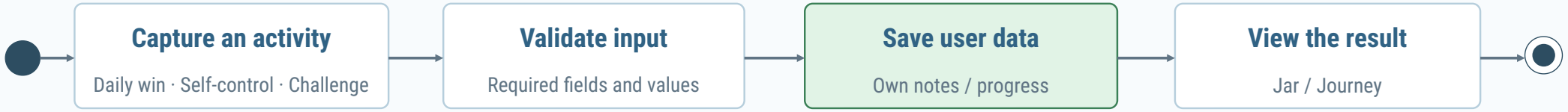
Streaks, statistics and local PDF

AI & Discord Assistance

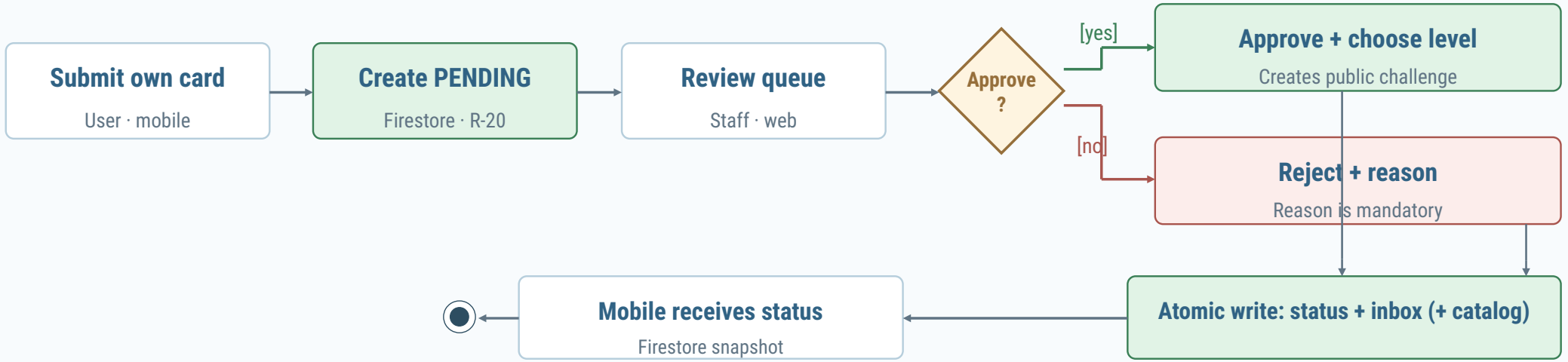
Gemini Free Tell and DM note management

Business Flow

A · DAILY ACTIVITY | USER / MOBILE

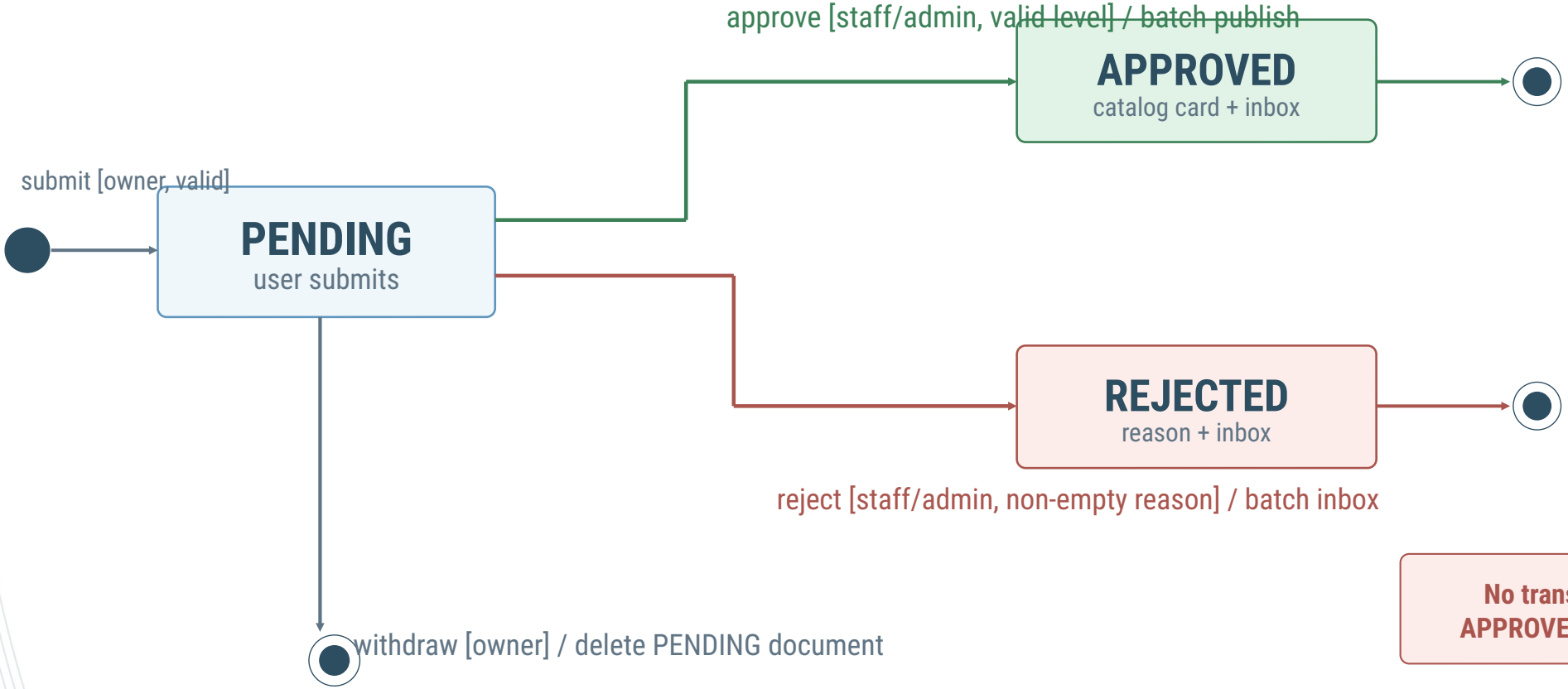


B · COMMUNITY CARD MODERATION | USER → STAFF / ADMIN → USER



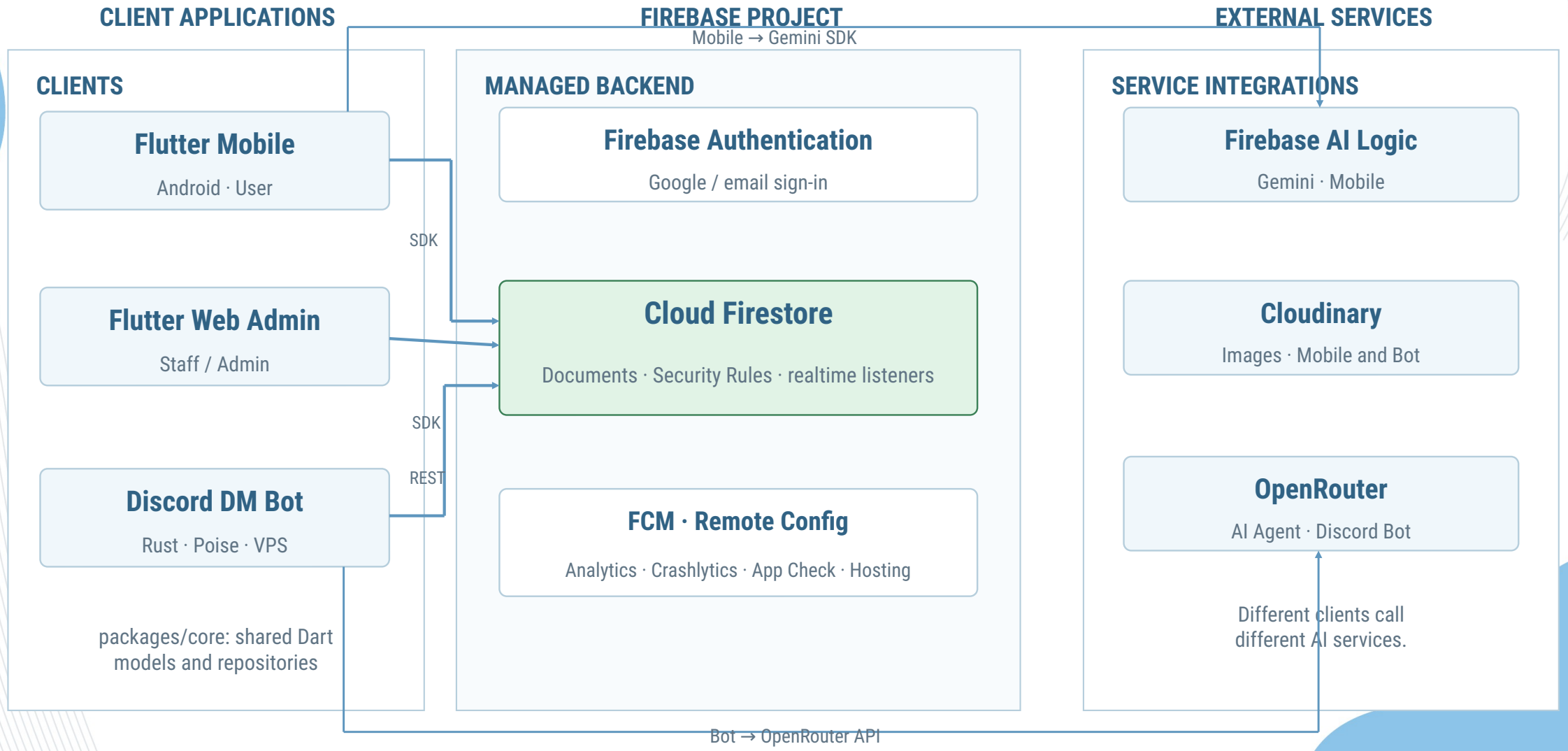
State Machine

ChallengeSubmission.status



No transition out of APPROVED / REJECTED

System Architecture



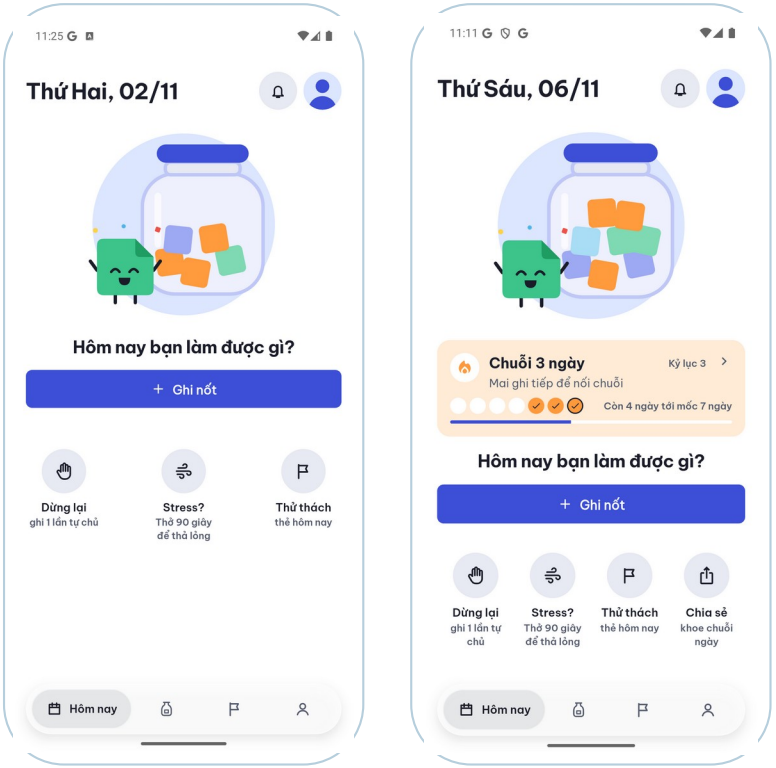
Technology Stack

Technology · What it is · Its role in Nốt

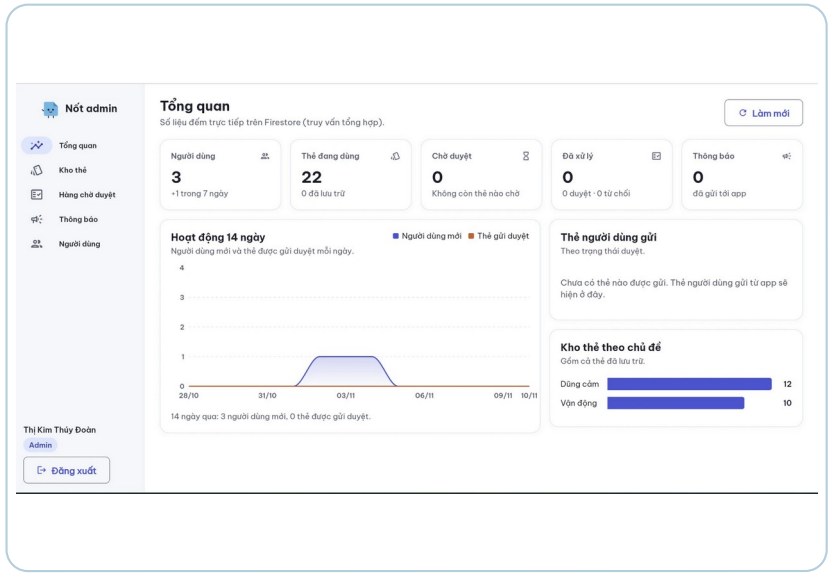
TECHNOLOGY	WHAT IT IS	IN NỐT
Flutter · Riverpod	UI framework · state management	Mobile and Web Admin interfaces
Firebase Authentication	Managed sign-in and user identity	Google/email login; UID-based access
Cloud Firestore	Realtime NoSQL document database	Notes, submissions, catalog and inbox
Firebase AI Logic · Gemini	Generative AI SDK and model access	Free Tell → editable note suggestions
FCM · Remote Config	Push messaging · remote feature flags	Notifications and feature switches
Analytics · Crashlytics	Event tracking · crash reporting	Usage evidence and error diagnostics
Cloudinary	Image storage and delivery service	Note, group and profile photos
Rust/Poise · OpenRouter	Discord bot · LLM API provider	AI note management via Discord DM
PDF · share_plus	PDF generation · native share sheet	Weekly card saved on the device
Hosting · GitHub Actions	Web hosting · CI/CD automation	Admin deployment and build checks

UI/UX Design

Same design system across Flutter Mobile and Web Admin



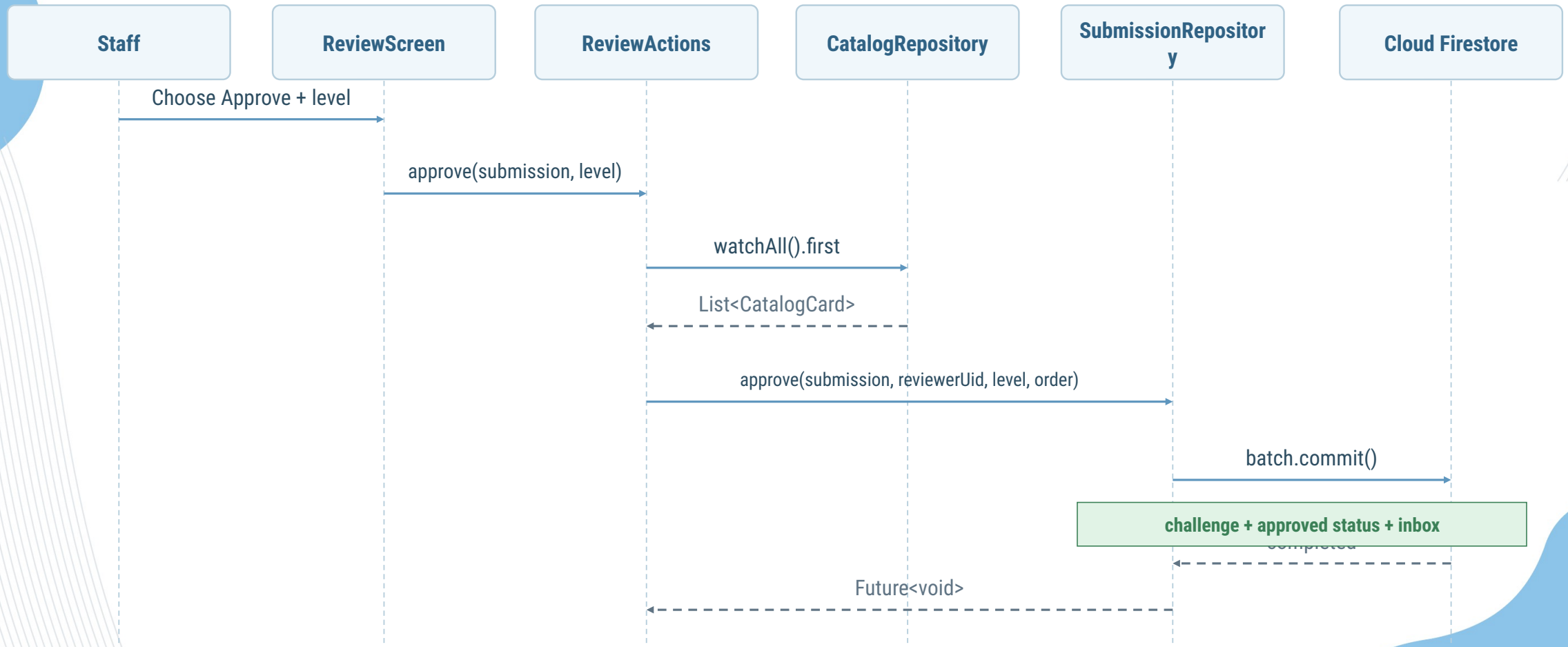
Mobile: Today · Jar · Challenges · Journey



Web: Dashboard · Catalog · Reviews
Announcements · Users

Sequence Diagram

Approve a community submission

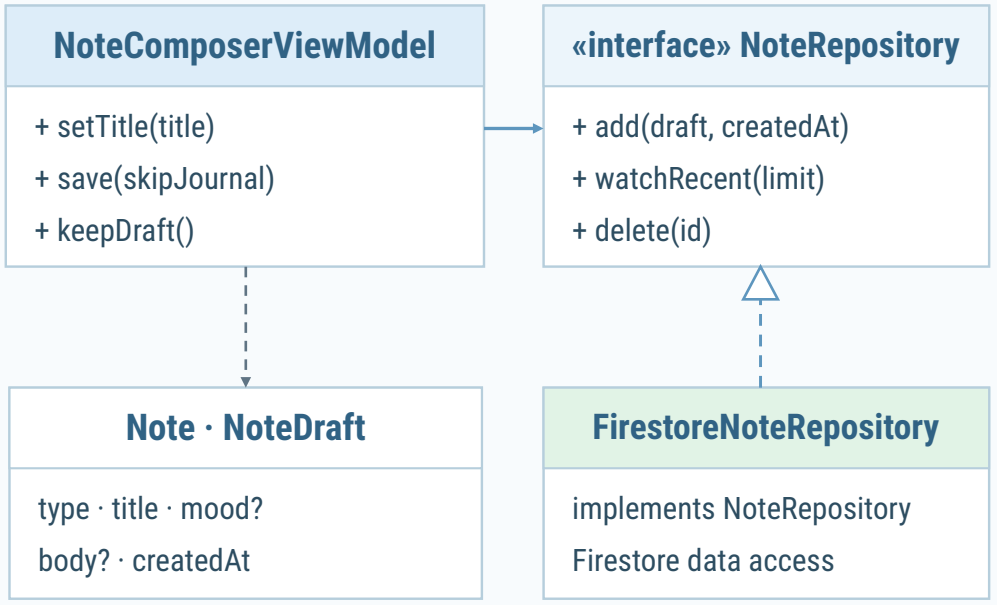


Precondition: status = pending; only Staff/Admin can review. Mobile receives Firestore snapshots.

Class Design

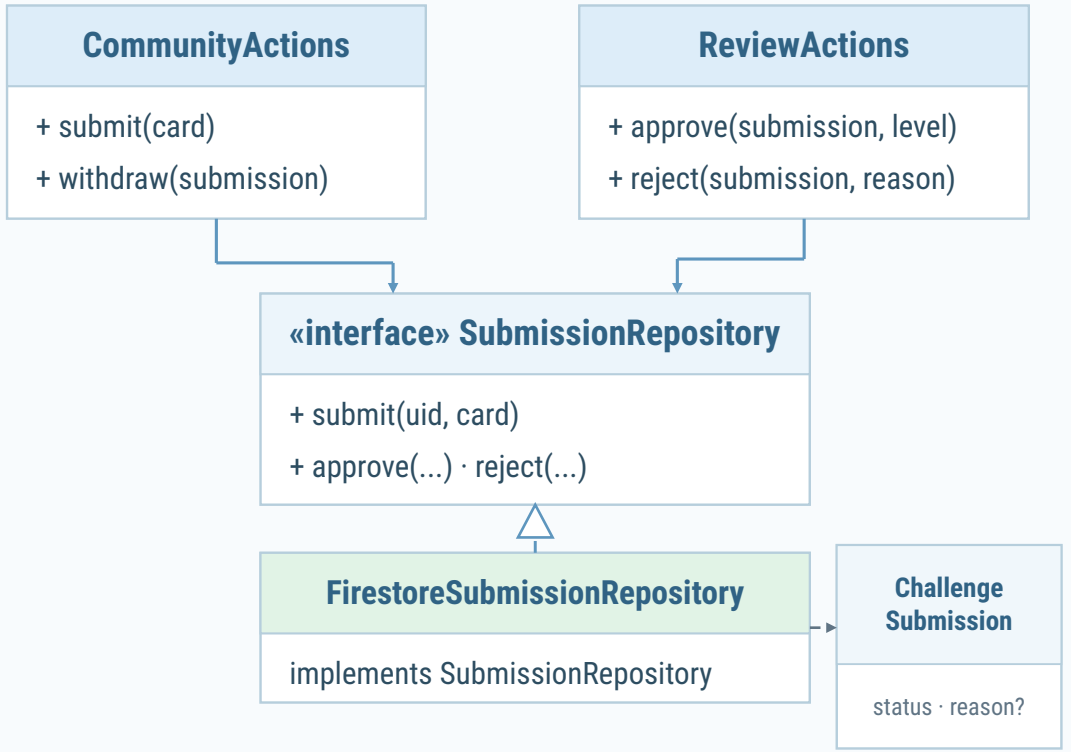
Representative classes across Mobile and Web Admin

DAILY NOTES · MOBILE



ViewModel depends on the interface.
Firestore implementation is injected through Riverpod.

COMMUNITY MODERATION · MOBILE + WEB

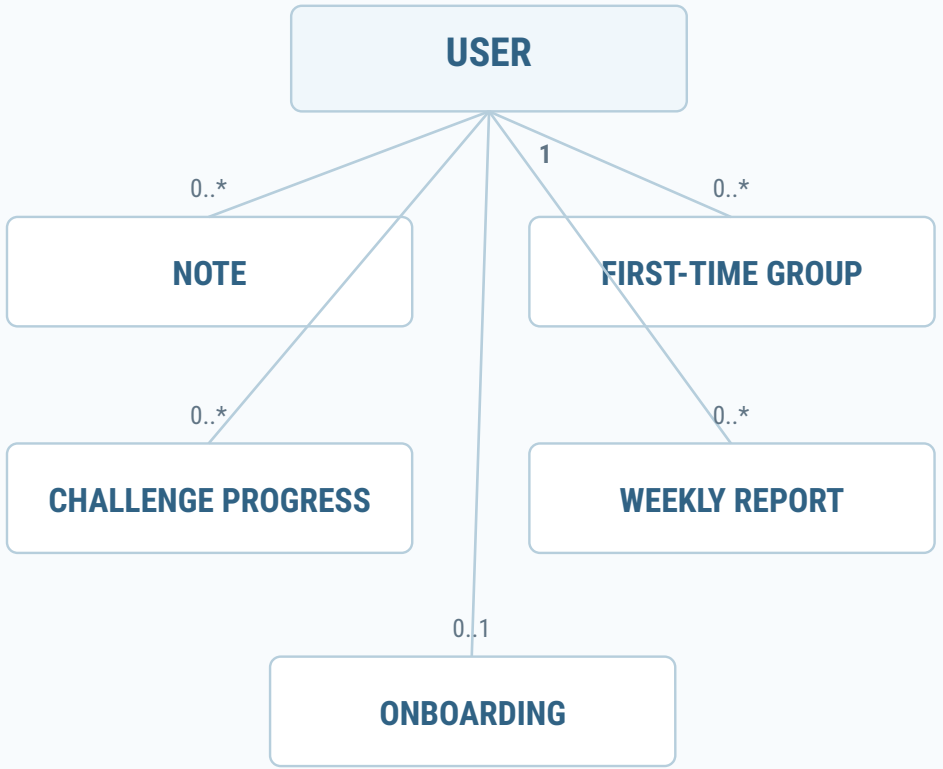


Role checks and state transitions: Firestore Rules.

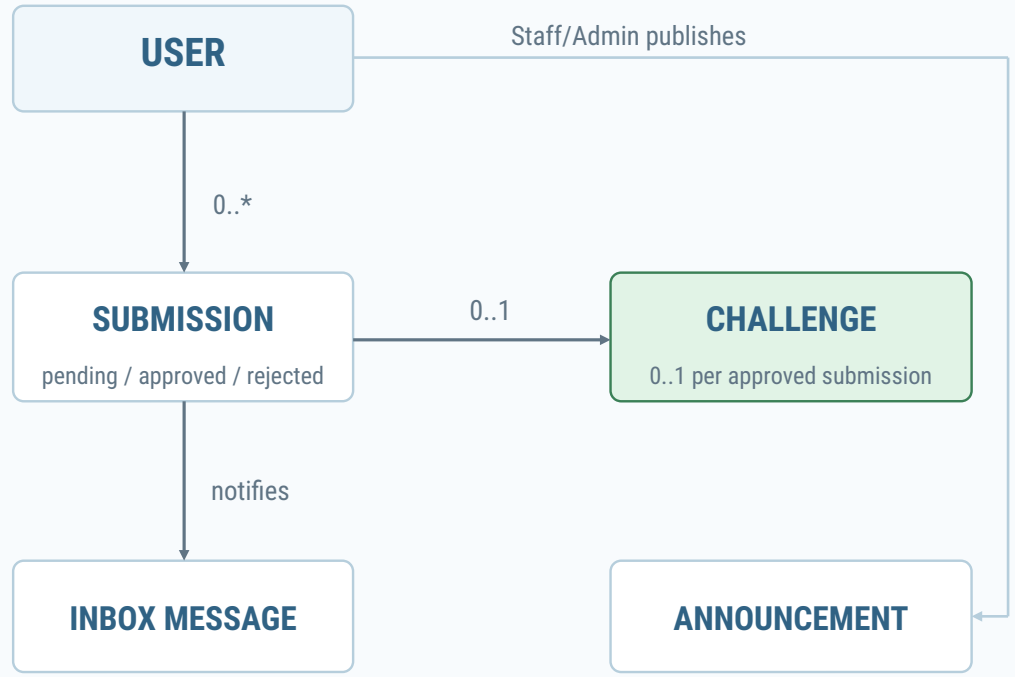
Database Design · Conceptual

Entities and their business relationships

PRIVATE USER DATA



SHARED & MODERATED DATA



Approved submission becomes a public challenge;
rejection only produces a reason and inbox message.

Database Design · Firestore

Firestore paths · key fields · access rules

USER-OWNED DOCUMENTS

users/{uid}
role · displayName · Self; admin can change others' roles

…/notes/{id}
type · title · createdAt · Owner; title 1–120

…/groups/{id}
name · image? · Owner

…/challengeProgress/{id}
topic · level / counts · Owner

…/customChallenges/{id}
title · topic · minutes · Owner; private card

…/private/onboarding
answers · planSource · Owner; validated values

…/inbox/{id}
kind · read · submissionId · Staff creates; owner reads

…/reports/{isoWeek}
fileName · sizeBytes · savedAt · Owner; PDF stays on device

SHARED / STAFF DOCUMENTS

challenges/{id}

topic · level · status

Staff manages; signed-in reads

challengeSubmissions/{id}

uid · status · reason?

Owner sends PENDING; staff reviews

announcements/{id}

title · body · authorUid

Staff posts; signed-in reads

**R-20…R-24: pending → approved/rejected;
approval + inbox committed atomically**

Firestore Rules: deny by default · 28/28 documented Rules tests passed

Traceability

Business function → implementation → data → verification

FUNCTION / ACTOR	IMPLEMENTATION	FIRESTORE DATA	EVIDENCE
Daily Note · User	NoteComposerViewModel → NoteRepository	users/*/notes	T02 · T04–T06 · T13
AI Free Tell · User	FreeTellViewModel → AiService	notes (after confirmation)	Gemini phone evidence; no 3× E2E
Community Card · User	CommunityActions → SubmissionRepository	challengeSubmissions	R-20 · R-21 · R-23
Moderation · Staff	ReviewActions → SubmissionRepository	submissions + catalog + inbox	R-22 · R-24 · D-1…D-3
Weekly PDF · User	WeeklyReportViewModel → ReportService	users/*/reports (metadata)	R-28; device PDF evidence

Evidence is versioned: ARTEMIS 41/42 refers to Lab 3; Final AI and PDF need updated E2E runs.

Demo Plan

Two user journeys: personal reflection and community moderation

WF-01

Record a Daily Win

User · Mobile

Saved note in Jar

WF-02

Submit a Community Card

User · Mobile

Submission = PENDING

WF-03

Review a Submission

Staff · Web Admin

APPROVED / REJECTED + inbox

—

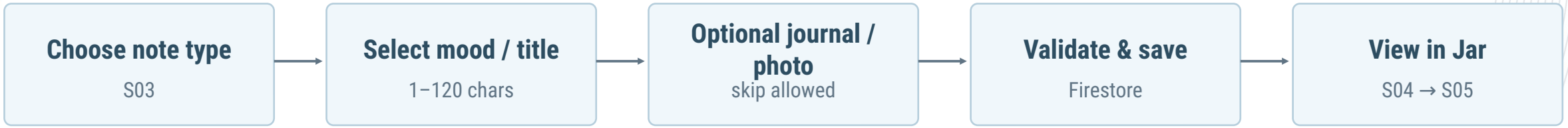
WF-01 is a separate daily workflow.

WF-02 output becomes WF-03 input.

Live proof: Mobile UI · Firestore status · Web review queue · User inbox

WF-01 · Record a Daily Win

Trigger: the user wants to record a positive moment.



BUSINESS RESULT

Saved note can be revisited in the Jar.

ALTERNATIVE PATH

Empty title blocked · Keep draft · Offline pending-sync.

Evidence: T02 · T04 · T05 · T06 · T13

WF-02 · Submit a Community Card

Trigger: the user wants to share a custom challenge.



BUSINESS RESULT

A pending submission appears in the review queue.

BUSINESS RULES

Owner-only · Valid fields · Flag-off hides Share.

Evidence: R-20 · R-21 · R-23 · evidence/cross-role/

WF-03 · Review a Community Card

Trigger: Staff/Admin opens a PENDING submission.



RESULT

Mobile shows decision; inbox receives a message.

EXCEPTION

Wrong role or empty rejection reason → denied.

Evidence: R-22 · R-24 · D-1 / D-2 / D-3

Testing & Verification

41 / 42

ARTEMIS E2E executions

Lab 3: 14 cases × 3 runs

28 / 28

Firestore Security Rules

R-01…R-28 · Emulator

D-1–D-3

Deployed cross-role trials

Three approvals: Web → Phone

T06 offline test: 1 agent timeout

The test did not establish an application defect.

D-2 ≤ 3.4 s · D-3 ≤ 2.3 s · D-1 21–23 s after idle/recovery

Achievements

MOBILE APPLICATION

Daily Notes · Challenges · Self-Control · Journey

WEB ADMIN

Role control · catalog management · submission review

LIVE CROSS-ROLE

Three deployed approval trials · inbox updates

AI & WEEKLY PDF

Gemini Free Tell on real phone · local PDF share

Evidence: mobile-final · web-admin · cross-role · Firestore Rules

Limitations & Next Steps

AREA	CURRENT LIMIT	NEXT STEP
User experience	<30 s check-in target not measured	Time real-user check-ins
E2E regression	Final PDF T14 + auth T15/T16 not run	Run updated ARTEMIS cases
Cross-role latency	D-1 21–23 s outlier after idle	Check reconnect behavior

Planned role-play / after-challenge AI is not counted as a completed feature.

Conclusion

Daily reflection, supported by real workflows.

- **PRODUCT** — Daily notes, challenges, and weekly reflection
- **SYSTEM** — Flutter Mobile + Web Admin + Firestore
- 28/28 Rules · 41/42 Lab 3 E2E · 3 deployed trials

github.com/FU-PRM-Projects/Lab-03 · prm393-not-lab3.web.app/admin/

Thank You

Questions?

Appendix A1 · Security Rules

Key authorization and workflow constraints

R-20	**User creates only their own pending submission**	Security Rules
R-22	**Only staff/admin can approve or reject; reason required**	state transition
R-23	**Owner withdraws only a pending submission**	Security Rules
R-24	**Staff creates inbox item; owner marks read**	Security Rules
R-26	**Admin changes role of others, never their own**	Security Rules
R-28	**User writes their own weekly PDF metadata**	Security Rules

Appendix A2 · AI & Discord

Nốt AI · In-App

Firebase AI Logic · Gemini
Free Tell → structured suggestions
Review, edit, then save
Consent + manual fallback

Discord DM Agent

Rust · Poise · OpenRouter
Linked account + Firestore REST
create_note · list_recent_notes
search_notes · optional photo

Two separate AI integrations; neither is the Web Admin moderation workflow.

Appendix A3 · Test Evidence

TEST TYPE	ID	RESULT	NOTES
ARTEMIS · Lab 3	T01–T14	41 / 42 passed	T06 offline-agent timeout
Security Rules	R-01–R-28	28 / 28 passed	Firebase Emulator, 10/11
Live cross-role	D-1	21–23 s UI update	First run, after idle
Live cross-role	D-2	≤ 3.4 s UI update	Approved + inbox
Live cross-role	D-3	≤ 2.3 s UI update	Approved + inbox
Final regression	T14 / T15 / T16	Not completed	Local PDF + auth paths

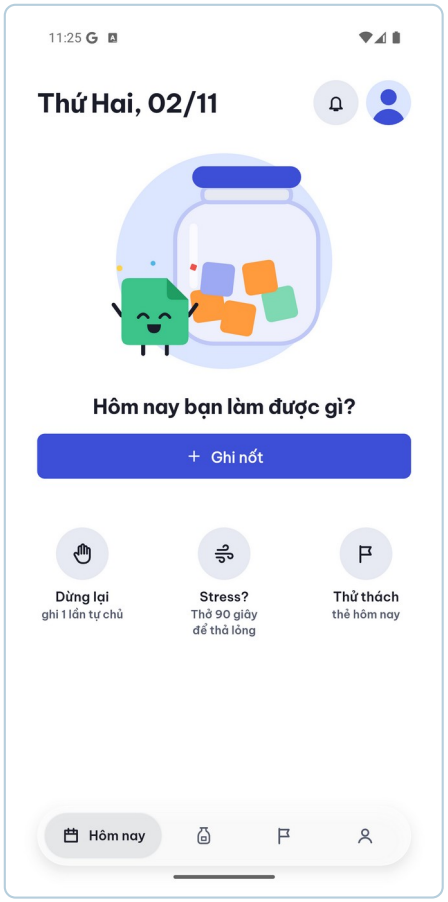
Appendix A4 · AI Code Review

52	3	37	1
CodeRabbit findings	Confirmed fixes	True, not yet fixed	Incorrect finding

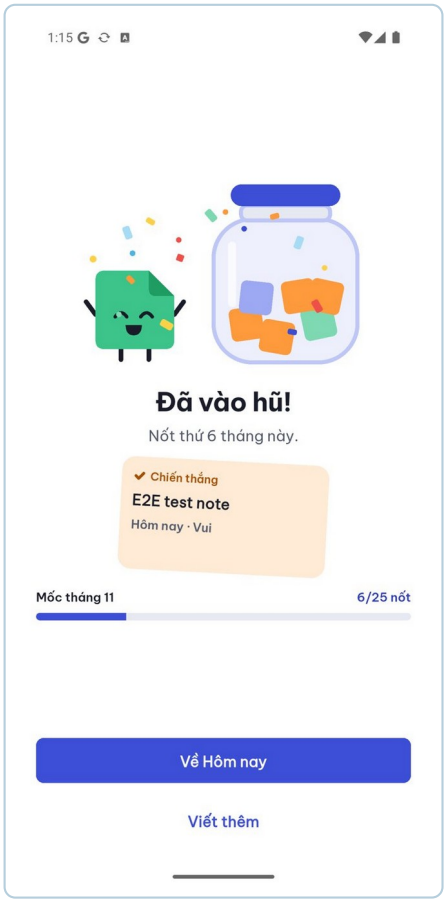
Examples fixed: clearing inbox on sign-out · notification routing · Analytics user ID

AI suggestions were reviewed by the team; 11 were not needed for the Lab scope.

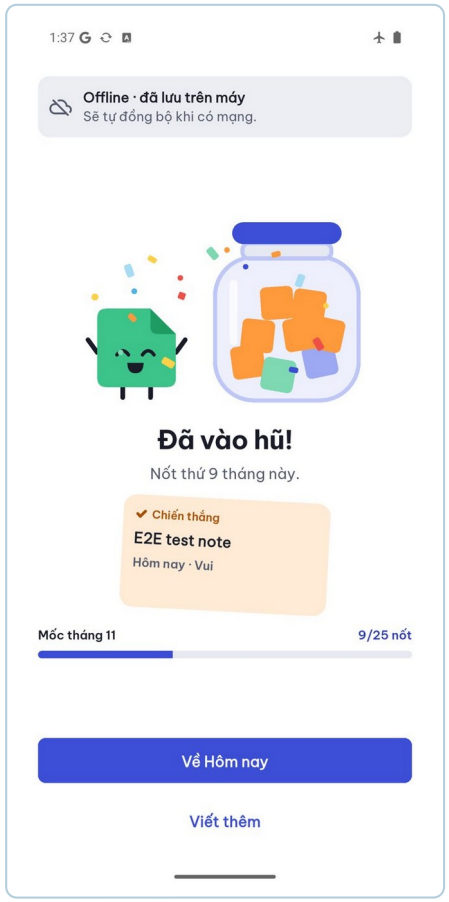
Appendix A5 · Mobile Evidence



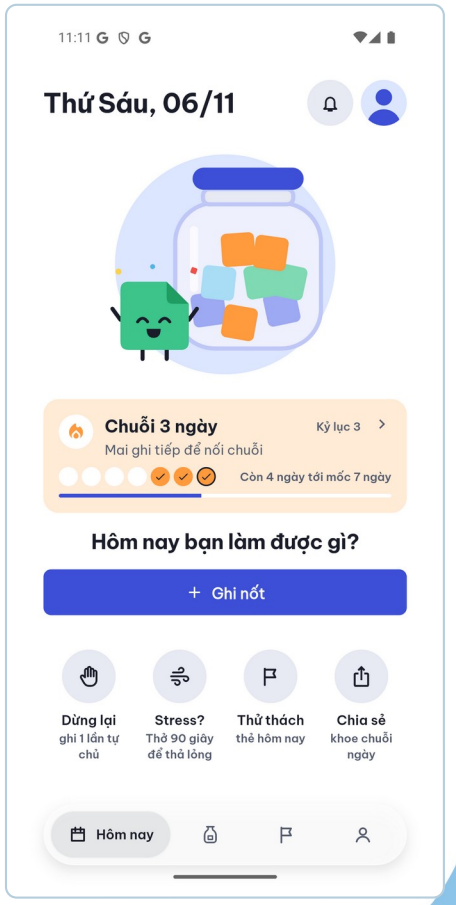
Today



Saved note



Offline case



Streak

Appendix A6 · Evidence & Scope

CONFIRMED	Code in main · role rules · cross-role tests · deployed UI
HISTORICAL	ARTEMIS 41/42 and mobile snapshots are Lab 3 evidence
PENDING	Local PDF ARTEMIS rerun · email auth E2E · <30 s timed study

Repository reference: main @ eb04447 (11 Nov 2026).